

AMATH 582: HOMEWORK 5

JAXON TUGGLE

Applied Mathematics Department, University of Washington, Seattle, WA
jmtuggle@uw.edu

ABSTRACT. This report compares Fully Connected Networks (FCNs) and Convolutional Neural Networks (CNNs) for image classification using the FashionMNIST data set. Three FCN models with varying weight budgets were trained and evaluated, while four CNN models were also examined. The report found that CNNs outperformed FCNs in accuracy for all models even with a reduced weight budget of 10K. Despite CNNs taking significantly longer to train, the models achieved higher testing accuracies showing a superior ability to classify image data.

1. INTRODUCTION AND OVERVIEW

The FashionMNIST dataset consists of grayscale images representing various clothing items, and has been used previously to evaluate the classification processes of neural networks. This report extends this evaluation to the performance of Fully Connected Networks (FCNs) and Convolutional Neural Networks (CNNs) for image classification. The primary goal is to evaluate the effects of different network architectures and weight budgets on model performance, with a particular focus on accuracy and efficiency. The assignment begins with the design and training of an FCN model constrained to 100,000 weights (or a relative order of magnitude). Hyperparameter tuning is conducted using the Adam optimizer, ReLU activation functions, and cross-entropy loss to achieve a testing classification accuracy exceeding 88%. Following this, the number of weights in the FCN model is varied by reducing the weights to 50K and 200K to investigate how these weight budget changes influence the model's classification accuracy and generalization capabilities. The next focus is on the implementation and training of CNN models also constrained by weight budgets. A CNN with 100K weights is trained and its performance was compared with that of the FCN variants. Further budget alteration is performed on the CNN model to create variants with 50K, 20K, and 10K weight budgets [2]. The performance of all models is analyzed based on several key metrics including training loss, validation accuracy, testing accuracy, and training time. A comprehensive comparison is created to explore the trade-offs between network complexity, computational resources, and overall model performance. This provided insights into the practical implications of different neural network architectures.

2. THEORETICAL BACKGROUND

A Fully Connected Neural Network (FCN) consists of an input layer, one or more hidden layers, and an output layer, with each neuron connected to every neuron in the next layer. The network's neurons apply the activation function ReLU

$$\text{ReLU}(x) = \max(0, x),$$

which outputs the input if positive and zero otherwise. Training loss is a measure of how well the model's predictions align with the true labels during training. The network learns by minimizing a loss function, such as Cross-Entropy Loss

$$L(y, \hat{y}) = -(y \log \hat{y} + (1 - y) \log(\hat{y})),$$

where y_i is the true label and $\hat{y}_i$ is the predicted probability for class i . The loss is minimized during training to improve the model's performance. To update the weights, the Adam optimizer is used which combines momentum and adaptive learning rates, using update rules

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) \nabla_w T$$

and

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) (\nabla_w T)^2,$$

where m_t and v_t represent the first and second moments. For more information on FCN, refer to Report 4 with other optimizers and regularization techniques. Convolutional Neural Networks are widely used for image classification tasks [1]. They consist of convolutional layers, pooling layers, and fully connected layers. The convolutional layer applies learnable filters to the input image to capture spatial features. The convolution operation is given by:

$$F[m, n] = I = [m, n] \cdot w[m, n] = \sum_{i=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} I[i, j] w[m - i, n - j]$$

Padding adds extra pixels around the image to preserve spatial dimensions. Padding types include no padding and zero-padding. The kernel size refers to the filter's dimensions, typically 3×3 or 5×5 . The output size of a convolution is:

$$n^{[l]} = \frac{n^{[l]} + 2p - f}{s} + 1$$

where $n^{[l]}$ is the input size, f is the kernel size, p is padding, and s is stride. Pooling layers reduce spatial dimensions and computational load. Max Pooling reduces the spatial dimensions of the input by selecting the maximum value from each region of the input, thereby downsizing the feature map. FC layers are used at the end of the network to make decisions based on learned features. The outputs of FC layers are given by:

$$y = f\left(\sum_{i=1}^n w_i x_i + b\right)$$

where w_i is the weight, x_i is the input, b is the bias, and y is the output. Validation and testing accuracy is the percent of correct predictions made by the model compared to the total number of predictions. It is calculated by

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Predictions}} \times 100,$$

where correct predictions are those where the predicted class matches the true class label. The FCNs and CNNs are trained over multiple runs in which the average of the testing accuracy was taken by adding each test accuracy after training divided by the total number of runs completed for each FCN and CNN. The total training time is also calculated throughout each run.

3. ALGORITHM IMPLEMENTATION AND DEVELOPMENT

The following techniques were all implemented in Python software. The FashionMNIST data set, containing 60,000 training images and 10,000 test images, was loaded and normalized. The normalization ensured that pixel values were scaled to have a mean of 0.1307 and a standard deviation of 0.3081. The data set was divided into training, validation, and test sets. A split is performed to ensure that each class is proportionally represented in both the training (90%) and validation (10%) sets. This all follows from work in Report 4. The FCN was implemented with a structure of 2 fully connected layers, where the input images were flattened and passed through two hidden layers before being classified by the output layer. The training process utilized the Adam optimizer with a learning rate of 0.001, and the model was trained using cross-entropy loss. Each epoch out of 40 total began by feeding the training batches into the model, where the loss was calculated, and the gradients were backpropagated for parameter updates. The running loss was recorded at each epoch to track the training progress. After each epoch, the model was evaluated on the validation set to assess its performance. The validation accuracy was computed by comparing the predicted labels with the true labels, and the progress was displayed through a progress bar. The test accuracy was calculated after the training was complete by comparing the model's predictions to the actual test labels. This procedure was repeated for multiple runs with training times and validation accuracies being recorded. This process was repeated with varying numbers of nodes to reach the specific weight budget allotments.

The CNN was designed to classify images using convolutional layers to capture spatial hierarchies. The architecture included two convolutional layers, each followed by max-pooling layers to reduce the dimensions. The input was passed through these layers and ReLU activation was applied. The output was then flattened before being passed through fully connected layers for classification with a hidden layer of 25 nodes. Just as the FCN models, the CNN was trained using the Adam optimizer with cross-entropy loss, and the model was evaluated on both the validation and test sets. The CNN's training procedure mirrored that of the FCN, where training loss was computed, and backpropagation was performed to update the model parameters. The model's performance was assessed by calculating the accuracy on the validation set at the end of each epoch. After training, the test accuracy was computed by comparing the predicted labels with the true labels from the test set. This entire process was repeated for four different CNN models with the only variation being the number of input and output channels of the convolutional layers. The models were run five times and the results were recorded for analysis. Key performance metrics, including training loss, validation accuracy, test accuracy, and training time were tracked across multiple runs. Training times were measured for each model, and average test accuracy was computed.

4. COMPUTATIONAL RESULTS

The first FCN model was hyperparameter tuned with the parameters described in the section prior. The FCN model with a 100K weight budget was designed without dropout regularization or batch normalization. The training loss and validation plots did not exhibit any signs of overfitting for the model, and the desire to strictly compare FC layers with Convolutional Layers for FCN and CNN models influenced this decision. For this reason, the only parameter changed was the number of nodes within the two hidden layers of the FCN models (1).

Model	Activation	Optimizer	LR	Loss	Layers
FCN 100K	ReLU	Adam	0.001	Cross-Entropy	[110, 110]
FCN 50K	ReLU	Adam	0.001	Cross-Entropy	[58, 58]
FCN 200K	ReLU	Adam	0.001	Cross-Entropy	[200, 200]

TABLE 1. Comparison of Architecture for FCN Variants.

While the inclusion of more hyperparameters would likely increase the accuracy of the models created, tuning still generated an FCN baseline model that achieved average testing accuracy greater than 88% (Figure 1). FCN 50K and FCN 200K models were created and compared in terms of training loss and validation accuracy. All three models exhibited reasonable training loss plots and validation accuracy plots. Reducing the weight budget to 50K had a noticeable effect on the validation accuracy while the other two models performed similarly.

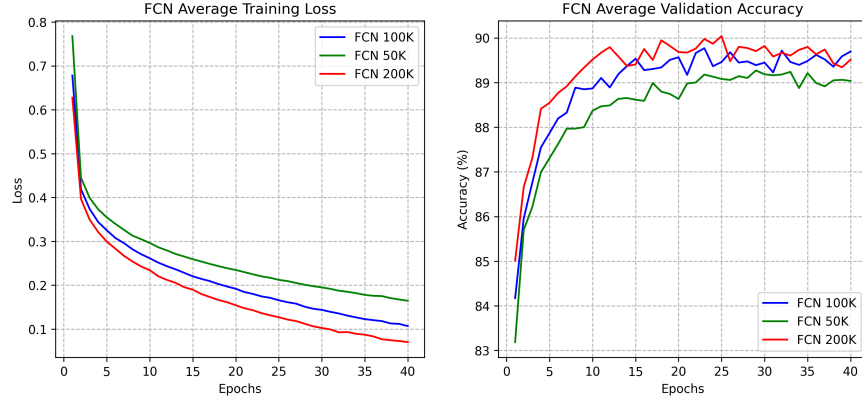


FIGURE 1. FCN model comparison for training loss and validation accuracy.

Next, the baseline CNN 100K weight budget model was created. Again, the number of hyperparameters was reduced to be able to clearly observe the effects of convolutional layers on testing accuracy. For this reason, the FC layer contained 25 nodes for all models as the input dimension changed based on the number of channels from the convolutional layers. This was to limit the variance between models and making the weight budget change solely on the number of convolutional layer input and output channels (Table 2).

Model	Activation	Optimizer	LR	Loss	Conv. Channels	FC Layers
CNN 100K	ReLU	Adam	0.001	Cross-Entropy	1→32→64	[3136, 25, 10]
CNN 50K	ReLU	Adam	0.001	Cross-Entropy	1→16→32	[1568, 25, 10]
CNN 20K	ReLU	Adam	0.001	Cross-Entropy	1→8→16	[784, 25, 10]
CNN 10K	ReLU	Adam	0.001	Cross-Entropy	1→4→8	[392, 25, 10]

TABLE 2. Comparison of Architecture for CNN Variants

The four CNN models exhibited reasonable training loss plots and achieved plateaus within their validation accuracy (Figure 2). The lowest training loss occurred with the highest weight budget and the highest validation accuracy. In fact, there was a noticeable decrease in validation accuracy as the weight budget was decreased for the CNN models, but the validation accuracy was higher for all CNNs in comparison to their FCN counterparts. The final step was to compare the test accuracy of all models with the training time. The highest average test accuracy occurred with the CNN 100K model (Table 3). This also had by far the longest training time. Varying the weight budget for the FCNs did not result in drastically different training times, as is evident for the CNN models. This indicates that changes in the number of nodes per layer does not effect training time while effecting testing accuracy. The CNN 10K model was comparable in testing accuracy with the FCN 100K and 200K models. This suggests that convolutional layers require less weight budget for similar accuracies with large FCNs. All CNNs took longer to train, so there is a trade for computational complexity and efficiency for higher testing accuracy.

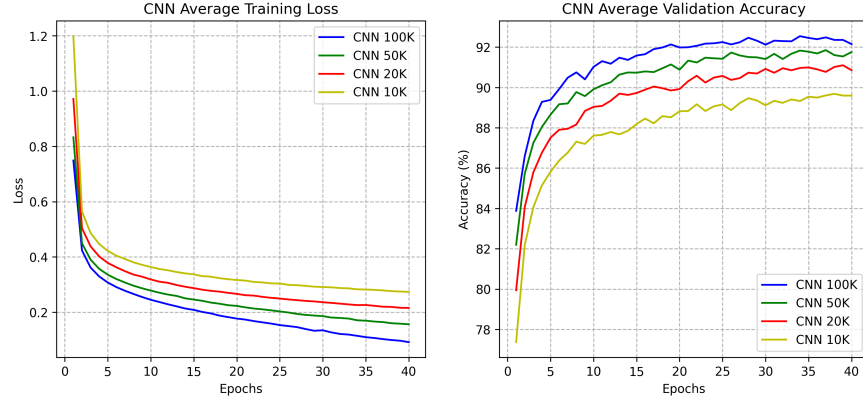


FIGURE 2. CNN model comparison for training loss and validation accuracy.

Model	Weights	Avg Testing Accuracy (%)	Training Time (secs)
FCN 100K	99,670	88.52 ± 0.38	208.88
FCN 50K	49,542	88.03 ± 0.37	201.25
FCN 200K	199,210	88.69 ± 0.30	209.93
CNN 100K	97,501	91.23 ± 0.42	1,298.99
CNN 50K	44,285	90.92 ± 0.22	761.18
CNN 20K	21,133	89.86 ± 0.40	517.45
CNN 10K	21,133	88.54 ± 0.26	388.51

TABLE 3. Comparison of FCN and CNN Variants

5. SUMMARY AND CONCLUSIONS

In this report, the performance of Fully Connected Networks (FCNs) and Convolutional Neural Networks (CNNs) were compared on the FashionMNIST data set. While FCNs achieved satisfactory testing accuracy, CNNs almost consistently outperformed the FCNs in terms of accuracy. The only similar testing accuracy occurred when the weight budget was reduced to 10K. However, CNNs required significantly more training time which presents a trade-off between computational complexity and performance. The results suggest that convolutional layers are more efficient at capturing relevant features with fewer weights than FCNs. Future study could explore different optimization techniques in CNN architectures as well as incorporating regularization techniques. Different models may also perform better on different data sets, so the models could be trained on various data to see classification accuracies.

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